

Assignment: Data Cleaning, Analysis & Visualization

Objective

The goal of this assignment is to develop practical skills in:

- Data Cleaning
- Data Analysis
- Outliers Handling
- Data Visualization

Students will work on a given dataset and apply real-world data science techniques.

Tasks to Perform

1. Data Cleaning

- Identify and handle **missing values**
- Remove or fix **duplicate records**
- Correct inconsistent data formats (e.g., dates, text cases)
- Convert data types where necessary

2. Data Analysis

- Display basic dataset information (head, info, describe)
- Perform **univariate analysis** (analyze each feature individually)
- Perform **bivariate analysis** (relationships between variables)
- Identify key patterns, trends, or anomalies

3. Outliers Handling

A. Graphical Methods

- Use:
 - Box plots
 - Histograms
- Identify outliers visually

B. Statistical Methods

- Use:
 - Z-score method
 - IQR (Interquartile Range) method
- Remove or treat outliers and justify your approach

4. Data Visualization (Comprehensive Task)

Students must create visualizations using **Pandas, Matplotlib, and Seaborn**.

A. Using Pandas Plotting Methods

Create the following plots:

- Area Plot
- Bar Plot
- Horizontal Bar Plot (barh)
- Line Plot
- Scatter Plot
- Histogram (hist)
- Density Plot / KDE (density, kde)
- Box Plot
- Hexbin Plot
- Pie Chart

B. Using Seaborn Visualization Techniques

1. Distribution Plots

- Distplot
- KDE Plot
- Rug Plot

2. Relationship Plots

- Joint Plot
- Pair Plot

3. Categorical Plots

- Factor Plot
- Box Plot
- Violin Plot
- Strip Plot
- Swarm Plot
- Bar Plot
- Count Plot

4. Matrix Plot

- Heatmap (Correlation Matrix)

Important Instruction

- If the provided dataset does **not support certain visualizations**:
 - Create your **own sample dataset**
 - Apply the remaining plots on that dataset

Submission Requirements

Submit the following:

1. Jupyter Notebook (.ipynb) or .pdf

- Well-structured and properly commented code
- All steps clearly explained
- All visualizations included

2. Summary Report

Include:

- Key insights from data
- Data cleaning steps performed
- Outliers handling explanation
- Final conclusions